

Joshua A. Duarte

Joshua.duarte151@gmail.com | joshuaduarte.com | (704) 960-5071

Education

University of California, Berkeley | Berkeley, California
Master of Engineering in Mechanical Engineering

Graduation: May 2023
GPA: 3.9/4.0

Texas A&M University | College Station, Texas
Bachelor of Science in Mechanical Engineering

Graduation: May 2022
GPA: 3.6/4.0

Professional Experience

Lucid Motors | Newark, California

Senior Mechanical Engineer, Drive Unit

April 2026 - Present

- Lead **Sustaining Engineering** efforts across Air, Gravity, and Sapphire, converting production and field issues into validated changes
- Reproduce failures through hands-on builds and testing, then driving corrections through validation and production release
- Prevent recurring defects by aligning designs with supplier capability, assembly variation, inspection strategy, and service needs

Mechanical Engineer II, Drive Unit

January 2024 - April 2026

- Designed and released drive unit components and interfaces for high-volume production using **CATIA V6**, **GD&T**, and DFM/DFA
- **Optimized datum schemes, tolerance stacks, and bolted joints** to improve assembly and manufacturing repeatability
- Performed overseas, in-person **supplier evaluation and selection** based on process capability, technical risk, and production readiness
- Revised differential components and interfaces to improve supplier capability, manufacturability, and production reliability
- Transformed initial concept into a production-ready design, coordinating overseas manufacturing to deliver on time and within budget

Boeing | Seattle, Washington

May 2021 - August 2021

Boeing Test & Evaluation Flight Test Design/Build Intern

- Designed, prototyped, manufactured, and installed **safety critical** foreign object debris shields on four Boeing 777X flight-test aircrafts
- Coordinated installation of test hardware used to vary aircraft center of gravity and electrical loads during flight testing

ExxonMobil | Houston, Texas

June 2020 - August 2020

UIS Wells Digital Innovation & Automation Intern

- Developed and enhanced a SharePoint platform to capture and distribute learnings found during the Well construction process
- Built data-entry workflows and interfaces using Nintex, JavaScript, jQuery, HTML, Excel, and VBA

Northrop Grumman | Palmdale, California

May 2019 - August 2019

Technical College Intern, Aerospace Systems, Top-Secret Clearance

- Assembled and tested mechanical, electrical, and electromechanical systems to verify build quality and consistency
- Created CATIA V5 models, assemblies, and drawings to support fabrication of aerospace hardware

Projects

Blue Goji | **Omni-Directional Treadmill**

- Advances an early-stage concept into a manufacturable **industrial design prototype** by refining requirements, simplifying geometry, and aligning the design with overseas supplier capabilities to minimize cost, weight, and improve efficiency
- Optimized parts for **injection molding, machining, and stamping**, resulting in a reduced system footprint by **57%**
- Simplified component geometry and assembly interfaces to reduce manufacturing complexity and improve build consistency
- Led **mechanical and software development** for a 10-person team, managing suppliers, BOMs, budgets, and prototype build schedules

Gambit | **Autonomous Chessboard**

- Designed and built an autonomous chessboard integrating a custom CoreXY gantry, mechanical systems, electronics, and software
- Iterated mechanism design and packaging through prototype builds to improve motion accuracy, reliability, and assembly fit
- Created custom components, assemblies, and fabrication drawings for final system manufacturing and integration

Aucosto | **Personal Operations Platform**

- Built a **full-stack** platform unifying planning, time tracking, finance, habits, projects, and task management
- Designed reusable dashboard widgets, shared event architecture, secure data flows, and modular service integrations
- Architected a cost-efficient **AI backend** using selective context retrieval, structured data processing, and optimized model calls

Leadership Experience

President | **Lucid Motors HOLA Employee Resource Group**

May 2025 - Present

- Oversee ERG operations, execution of initiatives, stakeholder engagement, and alignment with broader DEI and business objectives

President | **Society of Hispanic Professional Engineers**, College Station, Texas

May 2021 - May 2022

- Led 8 board members to continuously impact members through professional, academic, technical, and social development which resulted in more than 400 members registered with the organization, a **74.7%** increase in members from the previous year

Skills

- CAD (CATIA V6/SolidWorks/Fusion 360), Programming & Data Analysis, DFM/DFA, FEA, GD&T, Bilingual (English/Spanish)